

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm Examination

University Theory Examination October/November 2016

PH120 Pharmaceutical Engineering - II

Date: 09/11/2016

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION - I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION - I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION - I is 30 minutes. Answers to SECTION - I must be written during the first 30 minutes of the examination.
3. Answers to SECTION - II and SECTION - III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Clarification can be achieved by _____ and _____
[A] Filtration
[B] Centrifugation
[C] Both
[D] None
 2. The rate of filtration can be increase by increasing _____
[A] Surface Area
[B] Pressure
[C] Temperatue
[D] All
 3. What is the mechanism of surface filtration
[A] Straining
[B] Impingement
[C] Both
[D] None
 4. What is the pore size of membrain filter medium
[A] 100 μm
[B] 500 μm
[C] 1000 μm
[D] 10000 μm
 5. Which instrument is used to measure humidity?
[A] Thermometer
[B] Hygrometer
[C] Rotameter
[D] All
 6. Actual temperature measured by normal thermometer is known as _____
[A] Wet Bulb Temperature
[B] Dry Bulb Temperature
[C] Both
[D] None

7. Tray dryer work as _____ dryer
 - [A] Static bed
 - [B] Moving bed
 - [C] Fluid bed
 - [D] pneumatic
8. Spray drying is _____ process
 - [A] Batch
 - [B] Continuous
 - [C] Both
 - [D] None
9. Moisture sensitive drug can be processed in controlled _____
 - [A] Temperature
 - [B] Pressure
 - [C] Humidity
 - [D] All
10. Steam distillation required _____
 - [A] Hot water
 - [B] Water vapor
 - [C] Gas
 - [D] None
11. Molecular distillation is used for the distillation of _____
 - [A] Fixed oils
 - [B] Dibutyl phthalate
 - [C] Dimethyl Phthalate
 - [D] All
12. Which type of column is used in fractional distillation
 - [A] Plate
 - [B] Pack
 - [C] Both
 - [D] None
13. Water can be distilled by _____ distillation method
 - [A] Molecular
 - [B] Azotropic
 - [C] Simple
 - [D] Fractional
14. Starch product can be dry using _____ dryer
 - [A] Spray Dryer
 - [B] Freez Dryer
 - [C] Drum Dryer
 - [D] Fluid bed dryer
15. Freez drying can be occur at _____ temperature
 - [A] 15°C
 - [B] 10°C
 - [C] 0°C
 - [D] > 0°C
16. Percentage humidity is the ration of humidity to _____ humidity
 - [A] Unsaturated
 - [B] Saturated
 - [C] Stable humidity
 - [D] None

17. Which type of condenser is used in refrigeration system.
 [A] Water cooled
 [B] Air cooled
 [C] Evaporative condenser
 [D] All
18. Centrifugal effect is proportional to the _____ of the speed of rotation.
 [A] Square
 [B] Cube
 [C] Route
 [D] None
19. Which centrifuge is work on principal of sedimentation ?
 [A] Conical Disc Centrifuge
 [B] Tubular Bowl Centrifuge
 [C] Both
 [D] None
20. Which one is used as filter aid material?
 [A] Cellulose
 [B] Aluminium silicate
 [C] Keiselgul
 [D] All

SECTION – II

- Q 2** Attempt any TWO of the following
- | | | |
|----------|---|----|
| A | Describe the factor affecting evaporation. | 05 |
| B | What are the application of centrifugation in pharmacy? | 05 |
| C | Write the theory of drying in detail. | 05 |
- Q 3** Attempt any TWO of the following
- | | | |
|----------|--|----|
| A | Explain in detail the drying instrument used for thermolabile material drying. | 05 |
| B | Differentiate the azeotropic distillation and molecular distillation | 05 |
| C | Classify the evaporator used in pharmaceutical industry. | 05 |

SECTION – III

- Q 4** Attempt any FOUR of the following
- | | | |
|----------|--|----|
| A | Explain filter media, its ideal properties and factor affecting its selection. | 05 |
| | Give the examples of filter media. | |
| B | Explain in brief edge-filter. | 05 |
| C | Explain cartridge filter with suitable diagram. | 05 |
| D | Write a note on lyophilization. | 05 |

- A rotary drum filter turns at the rate of 2.0 rpm. The fraction of total filtering area immersed in the slurry is 0.20. It has been observed that 1.5 cu.ft. of filtrate is delivered per minute per sq.ft. of submerged area with given slurry under these operating conditions. If 6 ft³ of the filtrate is delivered per revolution by this rotary filter, what is the total area of the filter cloth of the drum?
- A filtration is carried out in a plate and frame filter press at constant pressure difference of 3.0 psi. the total filtering area is 80ft² and the sp. Cake resistance is 3.5 hr/ft. how many cubic feet of filtrate will be obtained in 1 hr if the cake contains 10Lb of dry solids per cubic feet of filtrate?
- *The resistance of the frame and cloth assumed as negligible and cake considered non compressible.
- Q 5 Attempt any FOUR of the following
- A Write a note on HVAC system. 05
- B Explain the psychometric chart. 05
- C Explain air conditioning. 05
- D Explain the effect of filter aids on mechanism of filtration. 05
- E Define: WBT and DBT with suitable examples. 05
- F Explain in detail about adiabatic condition 05